

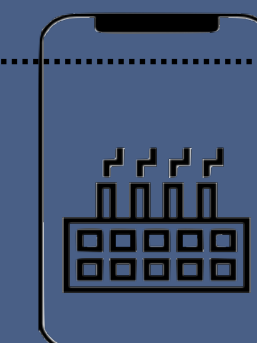
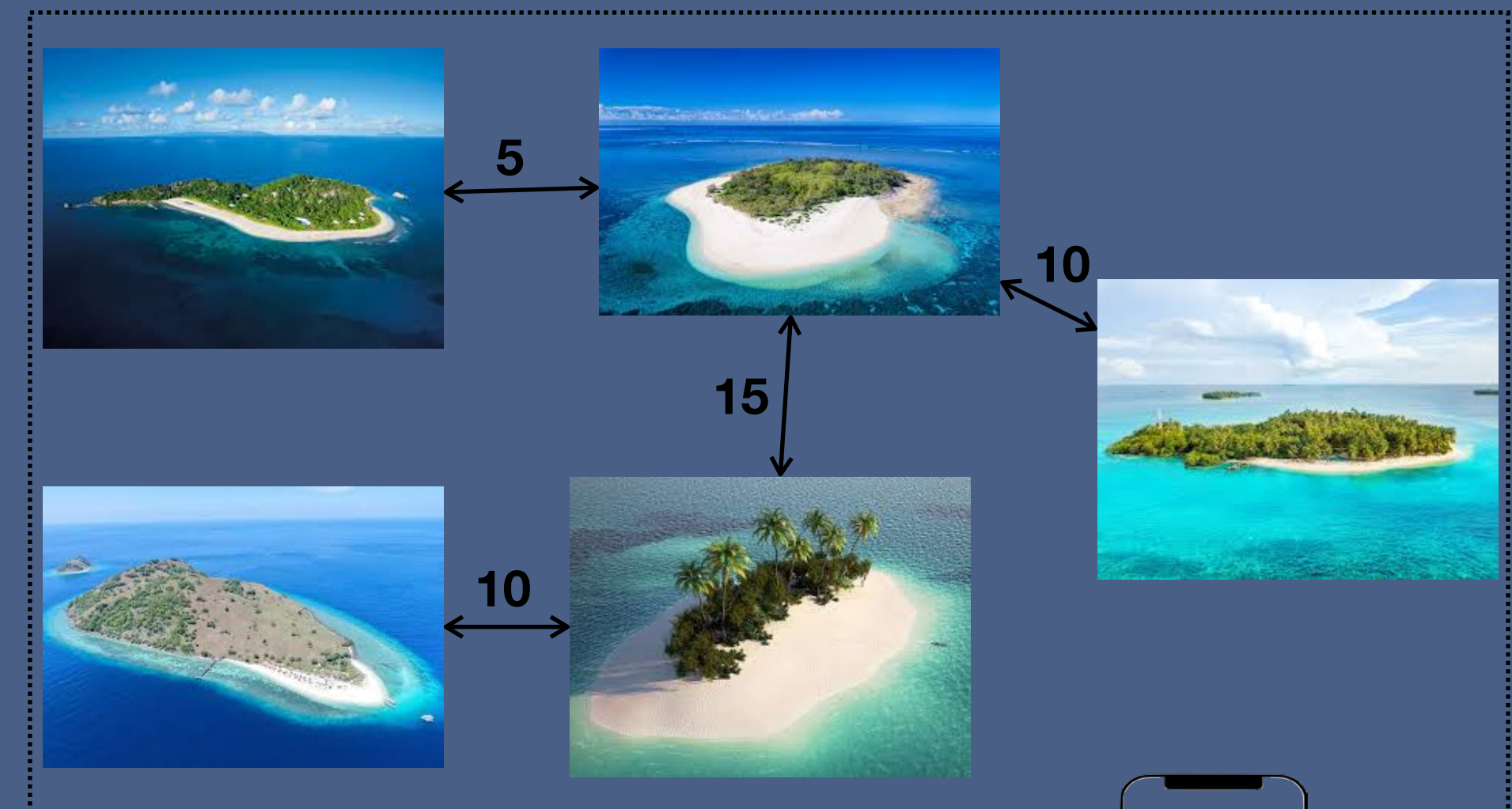
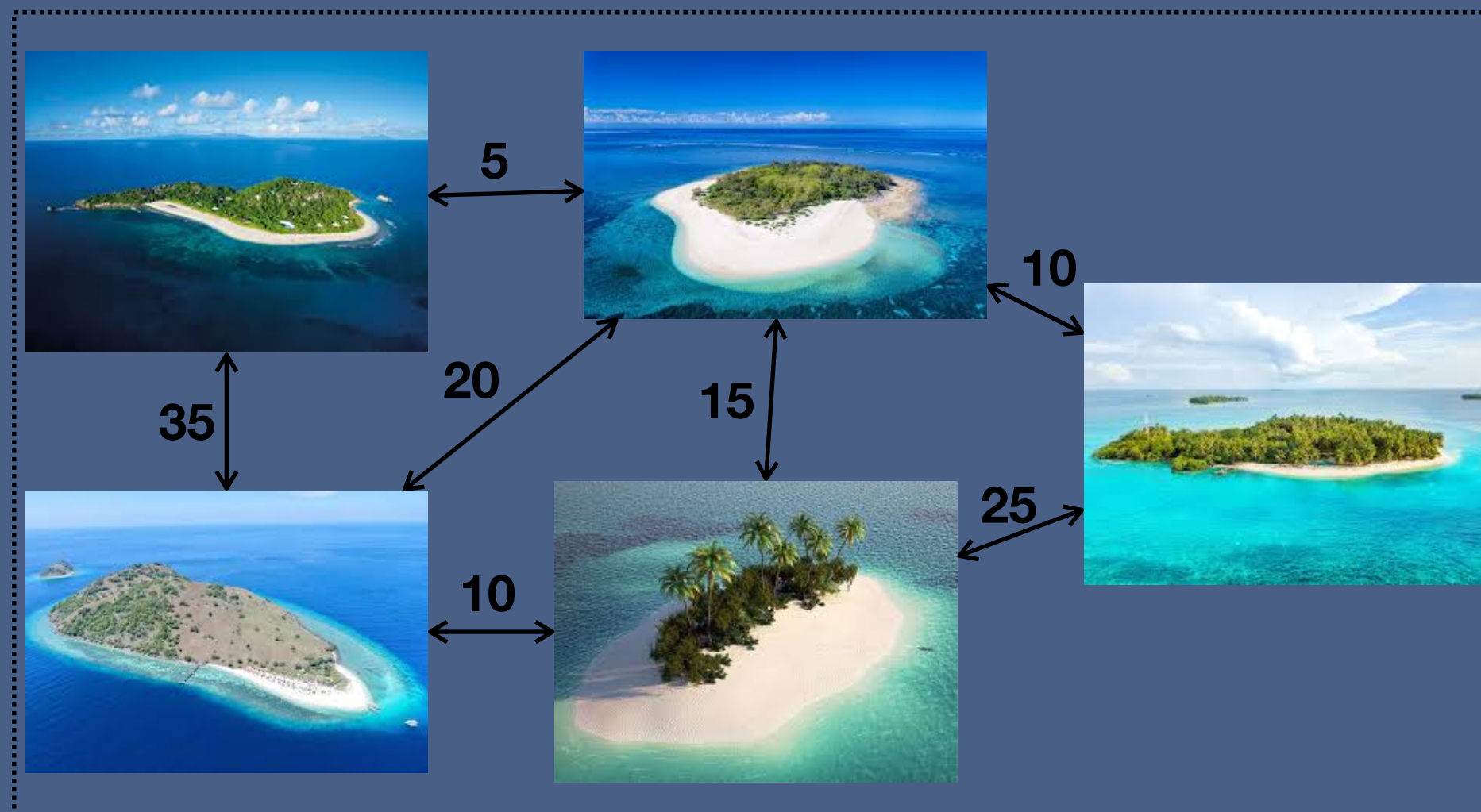
Minimum Spanning Tree

A Minimum Spanning Tree (MST) is a subset of the edges of connected, weighted and undirected graph which :

- Connects all vertices together
- No cycle
- Minimum total edge

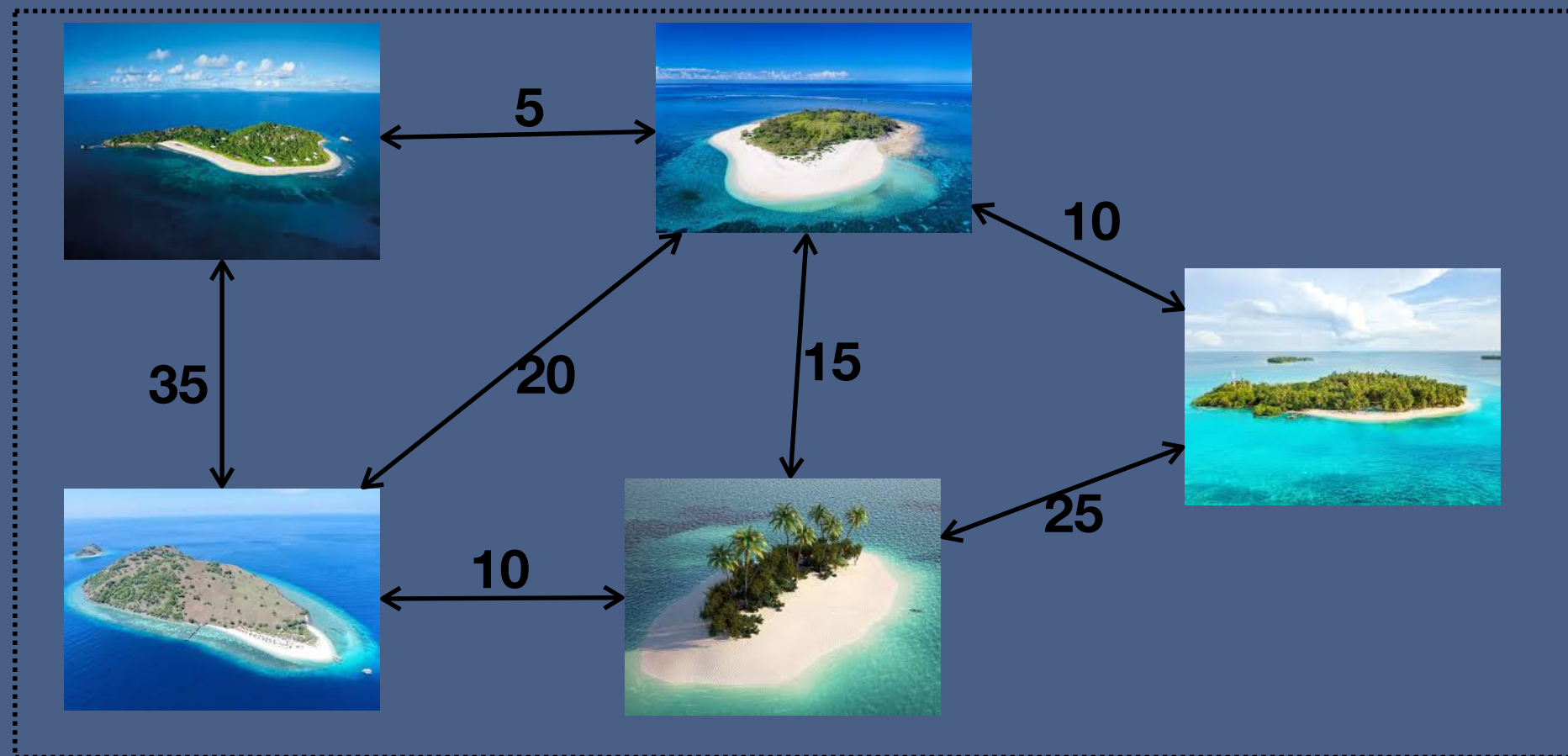
Real life problem

- Connect five island with bridges
- The cost of bridges between island varies based on different factors
- Which bridge should be constructed so that all islands are accessible and the cost is minimum

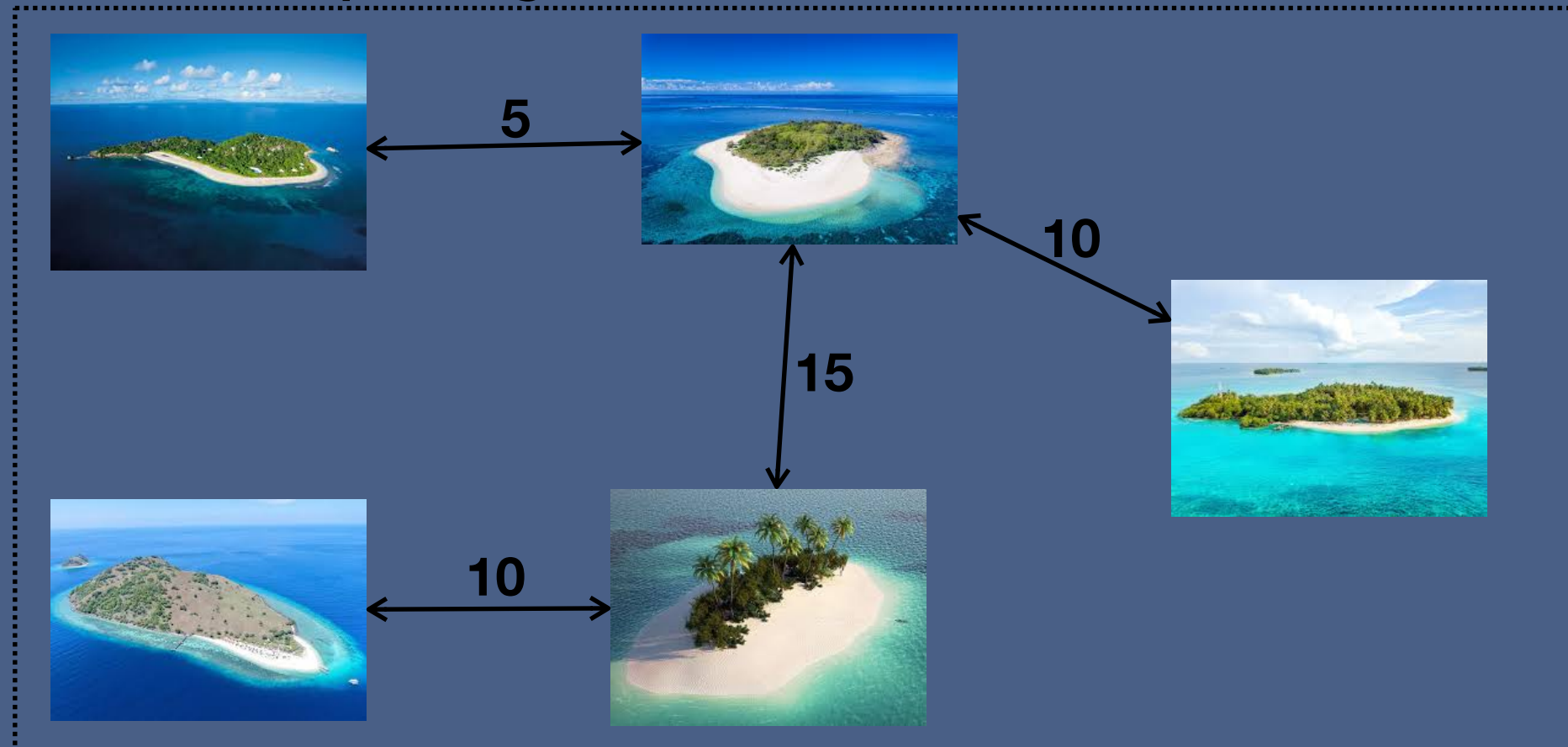


Minimum Spanning Tree

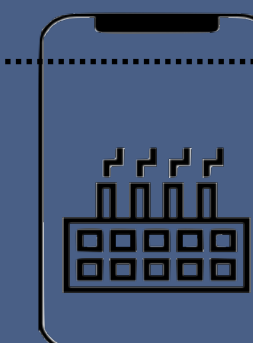
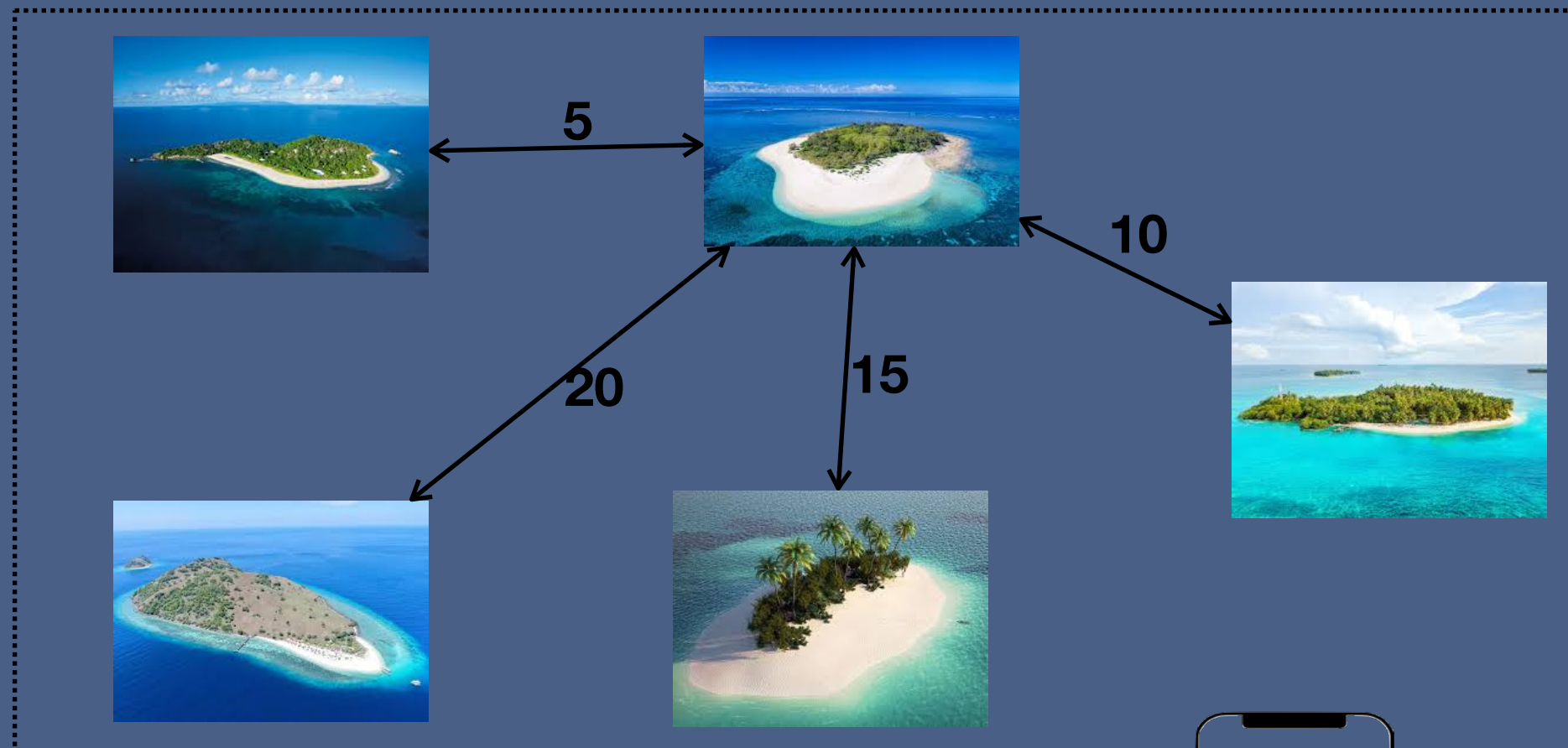
Real life problem



Minimum spanning Tree



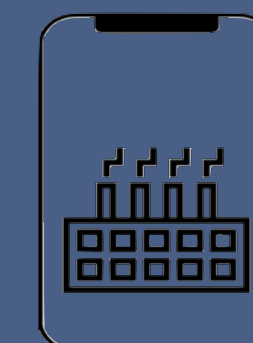
Single source shortest path



Disjoint Set

It is a data structure that keeps track of set of elements which are partitioned into a number of disjoint and non overlapping sets and each sets have representative which helps in identifying that sets.

- Make Set
- Union
- Find Set



Disjoint Set

makeSet(N) : used to create initial set

union(x,y): merge two given sets

findSet(x): returns the set name in which this element is there

